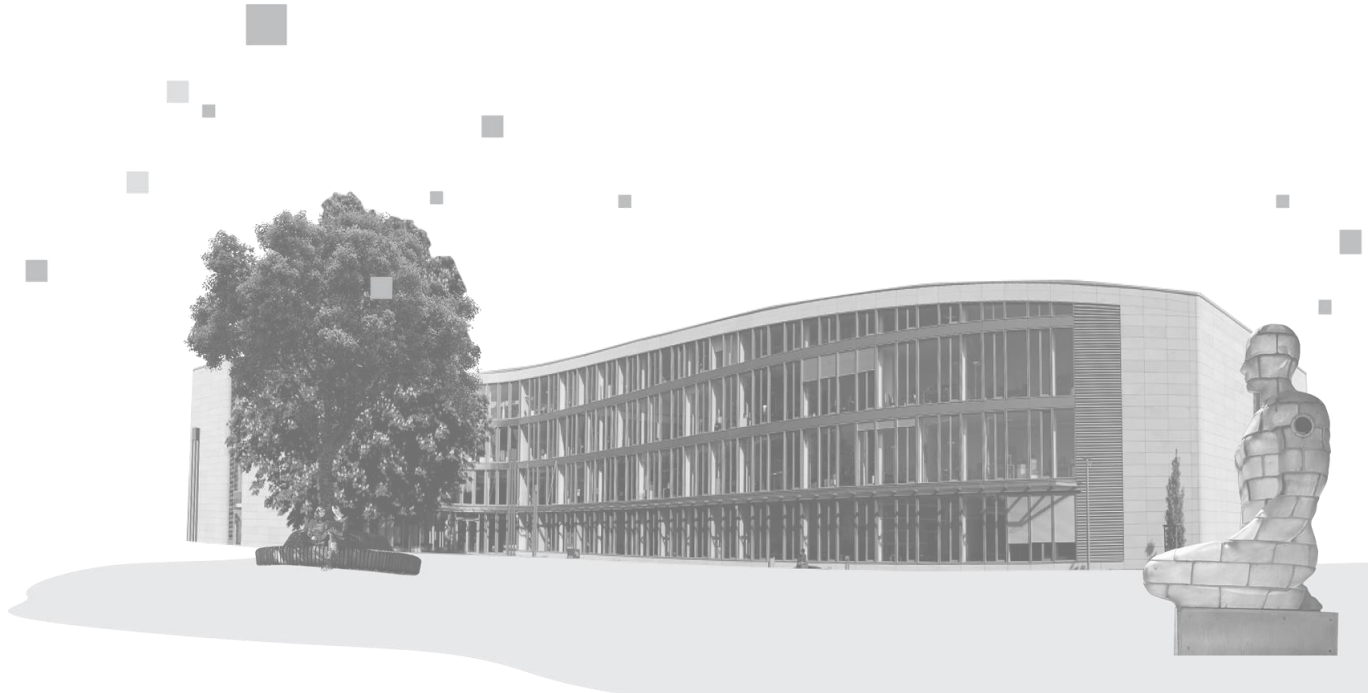
KI Service Zentrum
by Hasso-Plattner-Institut

HPI

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# AI Service Centre Berlin-Brandenburg



**Design IT.**
**Create Knowledge.**

www.hpi.de

# Guess the number

How many claims are in each text?

A The requirements must be documented so that they can be identified and taken into account during planning, operation, and procurement.

B Without binding password rules, there is a risk of weak or reused passwords that can be easily compromised.

C Issuance and revocation of entry, access, and usage media must be documented and replaced immediately if compromised.

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menti.com, code 7946 2006

# What is a claim?

| | |
|---|---|
| Everyday, whole | A claim is a sentence that states something. |
| Everyday, granular | Every clause that states something is a claim: main clause and subordinate clause count separately. |
| Logic | A claim is something that can be true or false on its own. Every element of a list counts for every action it belongs to. |
| Working definition | A claim is the smallest part of the reference answer that can be checked at one place in the Kompendium. |

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# Working definition

A claim is the smallest part of the reference answer that can be checked at one place in the Kompendium.

1. Every action (document, replace) is its own claim.
2. Every reason (because) and every purpose (so that) is its own claim.
3. A list that the Kompendium states as one list stays one claim.
4. Repetitions and filler do not count.

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# Example A: how many claims?

The requirements must be documented so that they can be identified and taken into account during planning, operation, and procurement.

| | | |
|---|---|---|
| Everyday, whole | the whole sentence | 1 |
| Everyday, granular | 1) The requirements must be documented<br>2) so that they can be identified and taken into account during planning, operation, and procurement | 2 |
| Logic | 1) The requirements must be documented.<br>2) Documented requirements can be identified.<br>3) Documented requirements can be taken into account during planning.<br>4) ... during operation.<br>5) ... during procurement. | 5 |
| Working definition | 1) Requirements must be documented.<br>2) Purpose: identify requirements.<br>3) Purpose: take into account during planning.<br>4) Purpose: during operation.<br>5) Purpose: during procurement. | 5 |

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# Example B: how many claims?

Without binding password rules, there is a risk of weak or reused passwords that can be easily compromised.

| | | |
|---|---|---|
| Everyday, whole | the whole sentence | 1 |
| Everyday, granular | 1) Without binding password rules, there is a risk of weak or reused passwords<br>2) that can be easily compromised | 2 |
| Logic | 1) Without binding password rules, there is a risk of weak passwords.<br>2) Without binding password rules, there is a risk of reused passwords.<br>3) Weak passwords can be easily compromised.<br>4) Reused passwords can be easily compromised. | 4 |
| Working definition | 1) Without password rules: risk of weak passwords.<br>2) Without password rules: risk of reused passwords.<br>3) Weak or reused passwords are easy to compromise. | 3 |

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# Example C: how many claims?

Issuance and revocation of entry, access, and usage media must be documented and replaced immediately if compromised.

| | | |
|---|---|---|
| Everyday, whole | the whole sentence | 1 |
| Everyday, granular | 1) Issuance and revocation of entry, access, and usage media must be documented<br>2) and replaced immediately if compromised | 2 |
| Logic | 1) Issuance of entry media must be documented.<br>2) ... access media ...<br>3) ... usage media ...<br>4) Revocation of entry media must be documented.<br>5) ... access media ...<br>6) ... usage media ...<br>7) Entry media must be replaced immediately if compromised.<br>8) Access media ...<br>9) Usage media ... | 9 |
| Working definition | 1) Issuance of entry, access and usage media must be documented.<br>2) Revocation of entry, access and usage media must be documented.<br>3) Entry, access and usage media must be replaced immediately if compromised. | 3 |

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# Guess the number, round 2

A claim is the smallest part of the reference answer that can be checked at one place in the Kompendium.

1. Every action (document, replace) is its own claim.
2. Every reason (because) and every purpose (so that) is its own claim.
3. A list that the Kompendium states as one list stays one claim.
4. Repetitions and filler do not count.

---

A The requirements must be documented so that they can be identified and taken into account during planning, operation, and procurement.

---

B Without binding password rules, there is a risk of weak or reused passwords that can be easily compromised.

---

C Issuance and revocation of entry, access, and usage media must be documented and replaced immediately if compromised.

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# What RAGAS tells the judge

The instruction to the judge, verbatim:

Given a context, and an answer, analyze each statement in the answer and classify if the statement can be attributed to the given context or not.

No definition. The judge decides for itself what a claim is.

Companies can apply for pilot projects and receive 2 to 3 days per week of active project support to develop a prototype together. They can also use the computing infrastructure.

gpt-oss-120b

## 2 claims

mostly splits per sentence

llama-3-3-70b

## 4 claims

splits finer

The same reference answer from notebook 2, section E b. The denominator of Context Recall depends on who splits.

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# RESEARCH

- **AI Methods Research**
- **AI Operations Research**



PubMedCLIP: How Much Does CLIP Benefit Visual Question Answering in the Medical Domain?

Sedigheh Eslami, Christoph Meinel, Gerard de Melo

Hasso Plattner Institu

{sedigheh.eslami, christoph

Abstract

Contrastive Language–Image Pre-training (CLIP) has shown remarkable success in learning with cross-modal supervision from extensive amounts of image–text pairs collected online. Thus far, the effectiveness of CLIP has been investigated primarily in general-domain multimodal problems. In this work, we evaluate the effectiveness of CLIP for the task of Medical Visual Question Answering (MedVQA). We present PubMedCLIP, a fine-tuned version of CLIP for the medical domain based on PubMed articles. Our experiments conducted on two MedVQA benchmark datasets illustrate that PubMedCLIP achieves superior results improving the overall accuracy up to 3% in comparison to the state-of-the-art Model-Agnostic Meta-Learning (MAML) networks pre-trained only on visual data. The PubMedCLIP model with different back-ends, the source code for pre-training them and reproducing our MedVQA pipeline is publicly available at https://github.com/sarahESL/PubMedCLIP.

1 Introduction

Medical visual question answering (MedVQA seeks answers to natural language questions abou a given medical image. The development of Med VQA has considerable potential to benefit health care systems, as it may aid clinicians in interpreting medical images and obtaining more accurate diag noses by consulting a second opinion. Thus, it has become a very active area of research, with compet itive benchmarks and yearly competitions (Abacha et al., 2021). Yet, visual question answering in the medical domain in particular remains non-trivial as we suffer from a general lack of large balanced training data, in part due to privacy concerns. To solve the multimodal task of MedVQA, a system must understand both medical images and textua questions and infer the associations between them sufficiently well to produce a correct answer (An

*Findings of the Association for Computati*
May 2-6, 2023 ©2023 Associ

Exploring Paracrawl for Document-level Neural Machine Translation

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Abstract

Document-level neural machine translation (NMT) has outperformed sentence-level NMT on a number of datasets. However, document-level NMT is still not widely adopted in real-world translation systems mainly due to the lack of large-scale general-domain training data for document-level NMT. We examine the effectiveness of using Paracrawl for learning document-level translation. Paracrawl is a large-scale parallel corpus crawled from the Internet and contains data from various domains. The official Paracrawl corpus was released as parallel sentences (extracted from parallel webpages) and therefore previous works only used Paracrawl for learning sentence-level translation. In this work, we extract parallel paragraphs from Paracrawl parallel webpages using automatic sentence alignments and we use the extracted parallel paragraphs as parallel documents for training document-level translation models. We show that document-level NMT models trained with only parallel paragraphs from Paracrawl can be used to translate real documents from TED, News and Europarl, outperforming sentence-level NMT models. We also perform a targeted pronoun evaluation and show that document-level models trained with Paracrawl data can help context-aware pronoun translation. We release our data and code here[1].

1 Introduction

The Transformer translation model (Vaswani et al., 2017), which performs sentence-level translation based on attention networks, has achieved great success and significantly improved the state-of-the-art in machine translation. Compared to sentence-level translation, document-level translation (Xu et al., 2021; Bao et al., 2021; Jauregi Unanue et al., 2020; Ma et al., 2020; Maruf et al., 2019; Tu et al., 2018; Maruf and Haffari, 2018) performs translation at document-level and can potentially fur-

[1]https://github.com/Yusser96/Exploring-Paracrawl-for-Document-level-Neural-Machine-Translation

*Proceedings of the 17th Conference of the European Chapter o*
May 2-6, 2023 ©2023 Associa

Efficient Parallelization Layouts for Large-Scale Distributed Model Training

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Abstract

Efficiently training large language models requires parallelizing across hundreds of hardware accelerators and invoking various compute and memory optimizations. When combined, many of these strategies have complex interactions regarding the final training efficiency. Prior work tackling this problem did not have access to the latest set of optimizations, such as FLASHATTENTION or sequence parallelism. In this work, we conduct a comprehensive ablation study of possible training configurations for large language models. We distill this large study into several key recommendations for the most efficient training. For instance, we find that using a micro-batch size of 1 usually enables the most efficient training layouts. Larger micro-batch sizes necessitate activation checkpointing or higher degrees of model parallelism and also lead to larger pipeline bubbles. Our most efficient configurations enable us to achieve state-of-the-art training efficiency results over a range of model sizes, most notably a Model FLOPs utilization of 70.5% when training a LLAMA 13B model.

1 Introduction

The number of parameters and computational resources spent on training deep neural networks is growing rapidly [1, 3, 14]. The largest models consisting of hundreds of billions of parameters do not even fit onto a single hardware accelerator. Thus, training these models requires various ways of reducing the memory requirements, such as ZeRO [16], activation checkpointing [2], and 3D-parallel (data, tensor, and pipeline parallel) training [13]. 3D parallelism, in particular, has been demonstrated to be effective for the training of Transformer-based large language models (LLMs) with hundreds of billions of parameters [13].

However, training these models efficiently with 3D parallelism requires significant domain expertise and extensive manual effort to determine the ideal configurations. These configurations not only need to combine data, model, and pipeline parallelism most efficiently, but also consider complex interactions with other memory and compute optimizations. FLASHATTENTION [5] in particular has had a notable impact since its release, enabling us to train models at previously impossible degrees of training efficiency. In light of these developments, we conduct a systematic study via a large-scale training efficiency sweep of these interactions. We consider up to 256 GPUs and LLAMA models with up to 65 billion parameters.

Workshop on Advancing Neural Network Training at 37th Conference on Neural Information Processing Systems (WANT@NeurIPS 2023).

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# EDUCATION

Newsletter

## Talks

tele-task.de/series/1463

- Guest talks on research and innovation





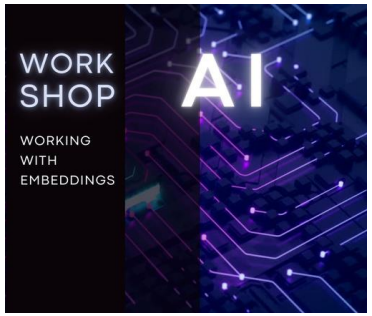





## Work shops

aimaker.community

- Practical topics
- Example topics: Speech2summary, Docker for ML, semantic search





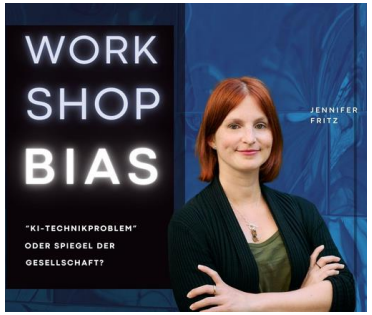


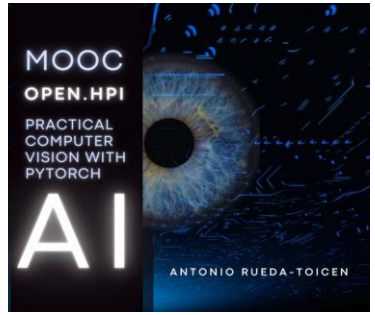



## MOOCs

open.hpi.de/channels/ai-service-center

- ChatGPT: What does generative AI mean for our society?
- Profitable AI
- Understanding and avoiding AI biases







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# INFRASTRUCTURE

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aisc.hpi.de







- Access **free of charge**
- No production operation
  - Data should be **anonymised** or **synthesised**
  - No **hosting of products**
- **Reporting** & **publication** by users
- **Existing rights** remain with users
- **New rights** remain with users
  - Granting of rights for research and teaching

**Training**
- 64 NVIDIA H100 GPU

**Inference**
- 40 NVIDIA A30 GPU

**ARM Server**
- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

**GPU Server**
- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

**Edge**
- ARMv8 CPU
- NVIDIA Jetson AGX Module

**Neuromorph**
- 288 SpiNNaker

**Storage**
- 1.5 PB NVRAM

**Network**
- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

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# ADVISORY

Consultation

## AI Consultation

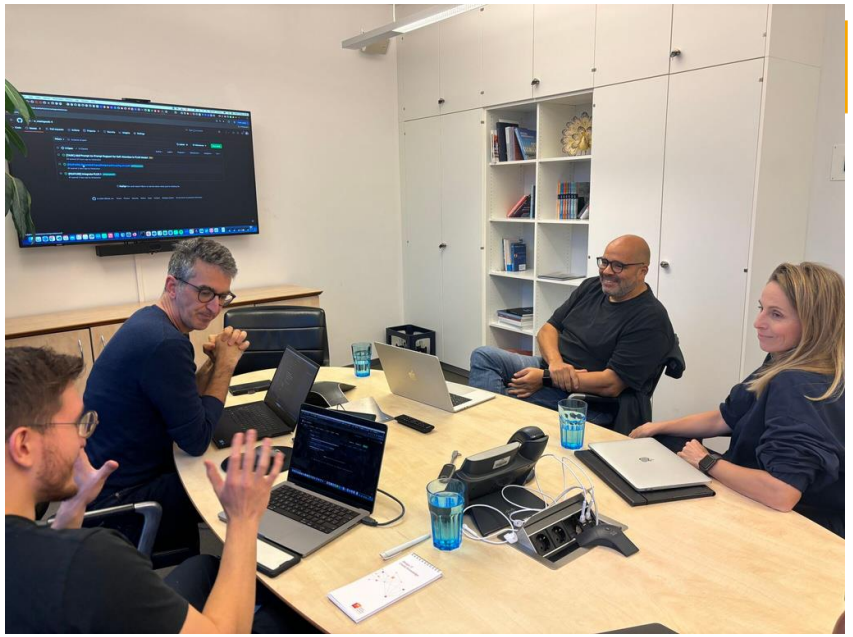
- AI Infrastructure
- AI Models & Frameworks
- AI Use Cases

## AI Pilot Projects

- Co-development of a prototype
- Applications every three months
- Selection criteria e.g. AI maturity, common good
- Publication of results

## Collaborations

- Organisied community meetings



**Previous AI Pilot Projects**

- Generation of maths problems
- Generation of plain language
- Generation of upcycling suggestions
- Reduction of food waste
- Dating of handwriting

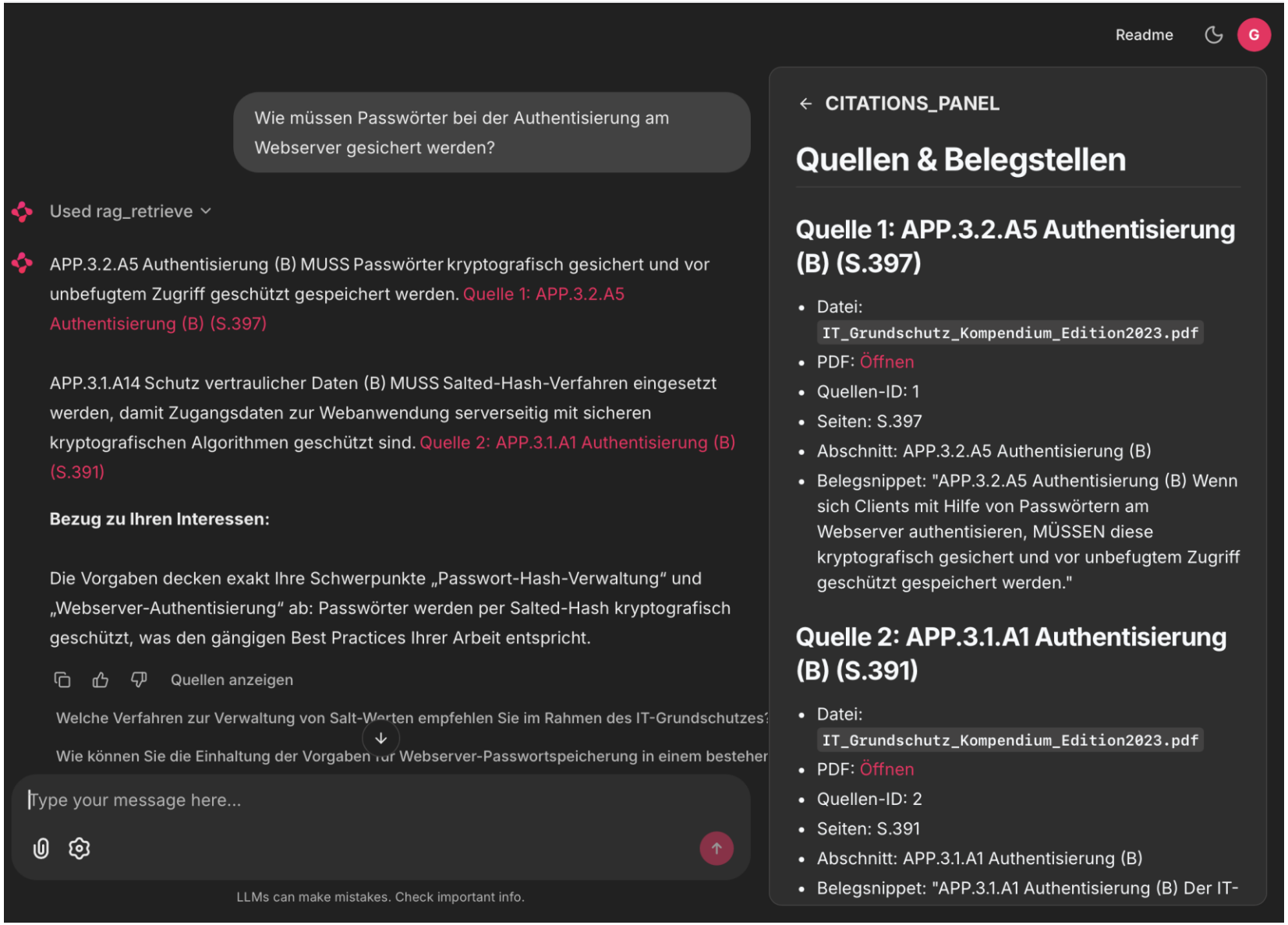
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# PILOT PROJECT: RAG SYSTEM

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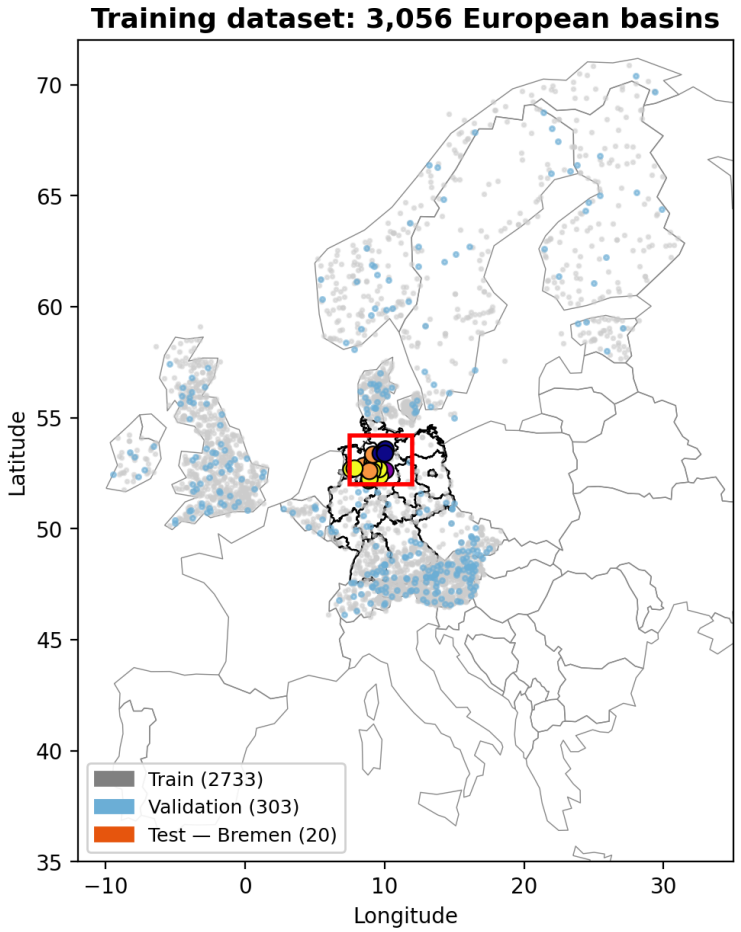


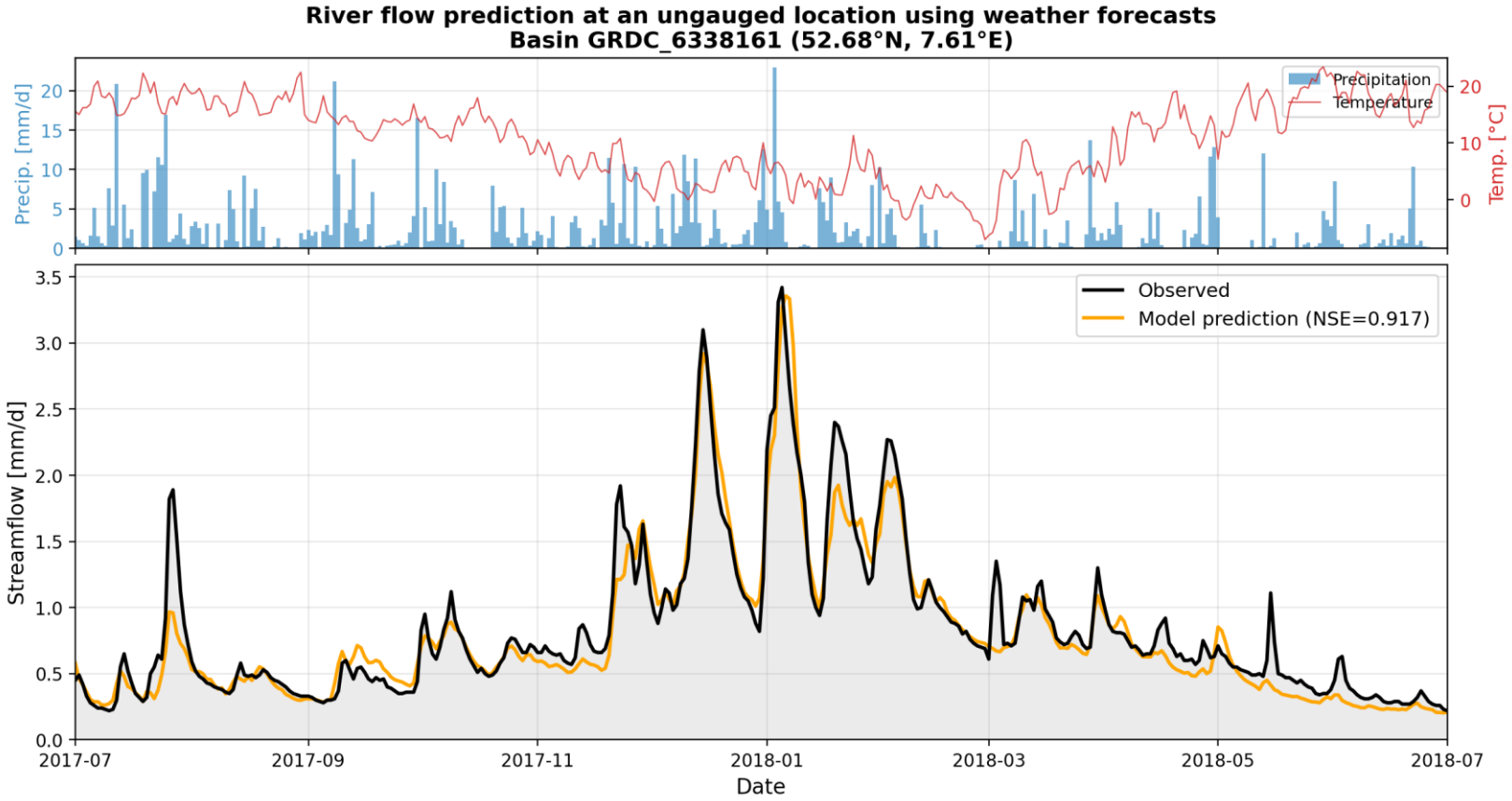

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# PILOT PROJECT: OPEN RIVER SENSE

KI Service Zentrum by Hasso-Plattner-Institut



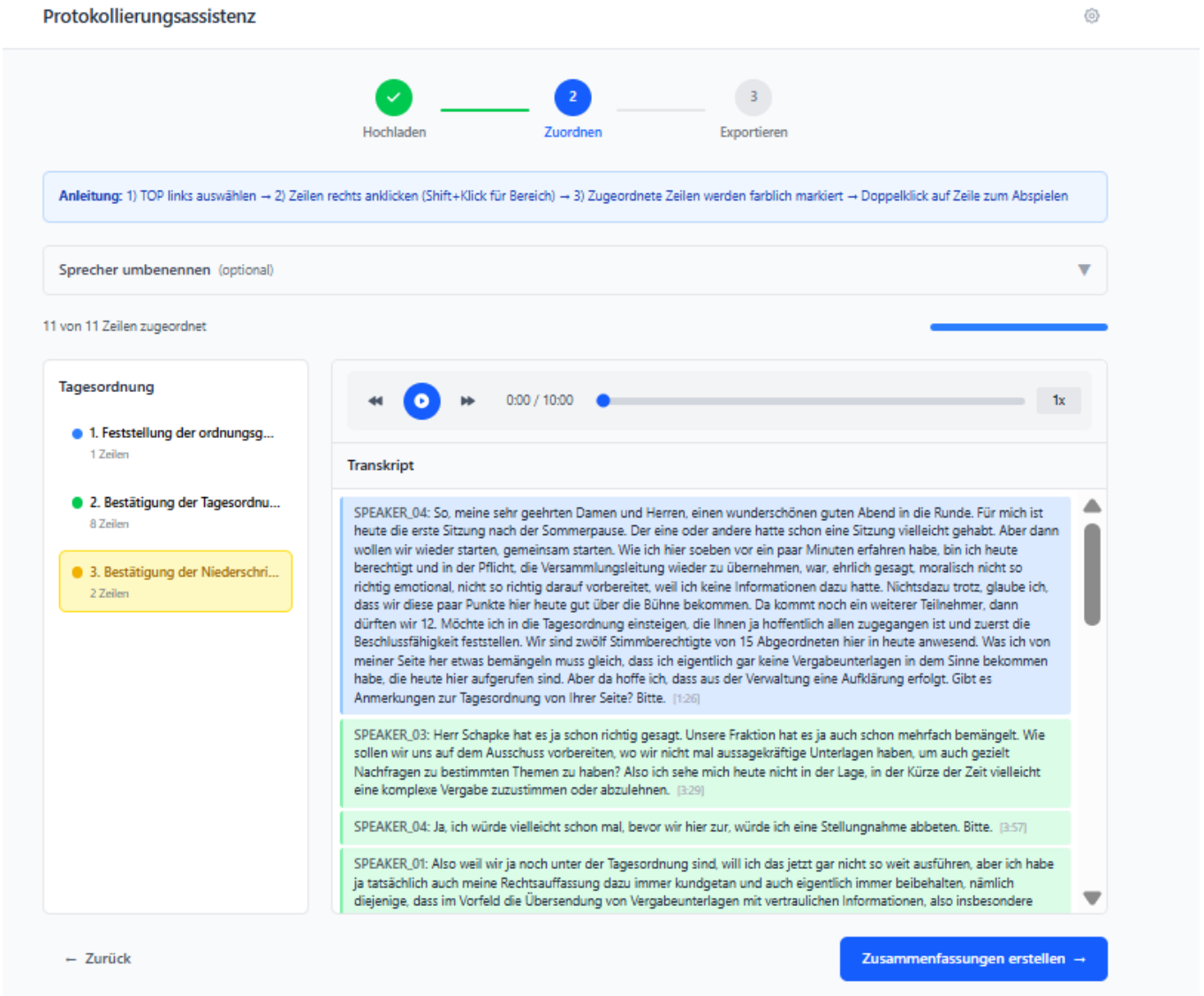




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# PILOT PROJECT: AUTOMATIC TRANSCRIPTION

KI Service Zentrum by Hasso-Plattner-Institut HPI




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# PILOT PROJECT: GENERATIVE AI

KI Service Zentrum by Hasso-Plattner-Institut

huggingface.co/datasets/aihpi/fashion-prompt-to-prompt



**A cropped jacket with patch pockets, featuring bold tribal prints and tassel closures.**

**Original Caption**

Add a detachable faux fur collar to the neckline for winter wear.

**Edit Instruction**



A cropped jacket with patch pockets, featuring bold tribal prints and tassel closures, **now includes a detachable faux fur collar at the neckline.**

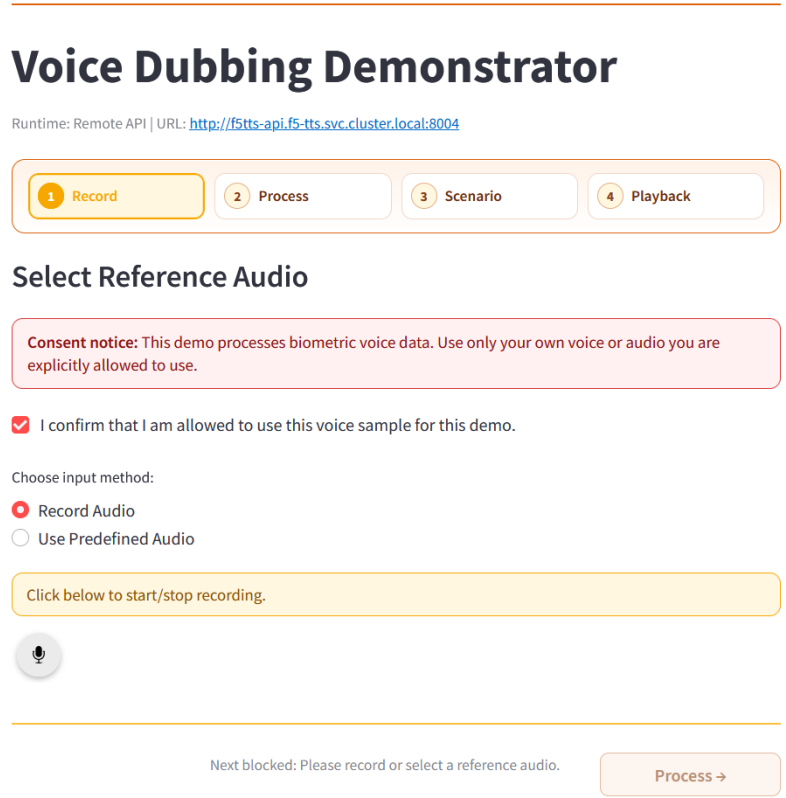
**Resulting Caption**

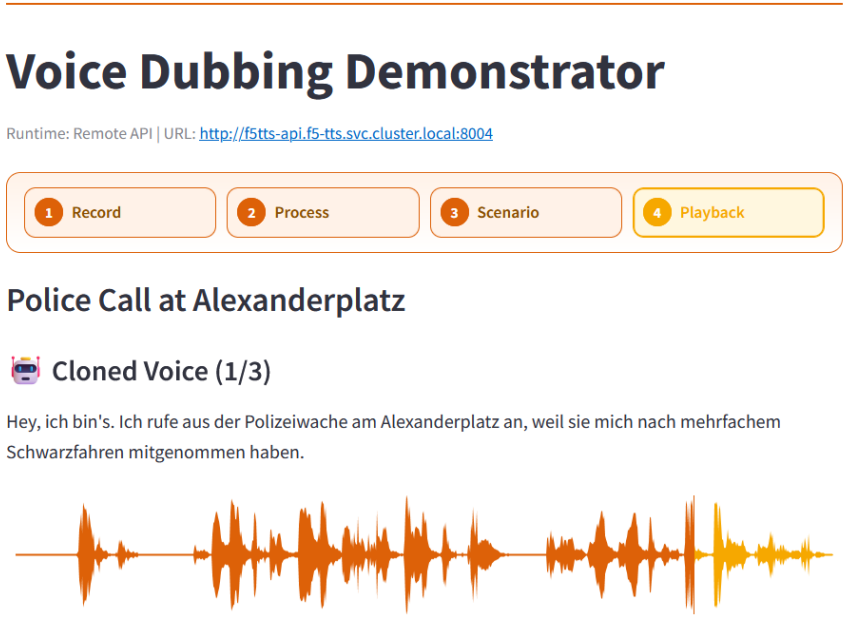
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# DEMONSTRATOR: VOICE CLONING

https://tts.aisc.hpi.de/
user: event
pw: aisc@hpi2022!

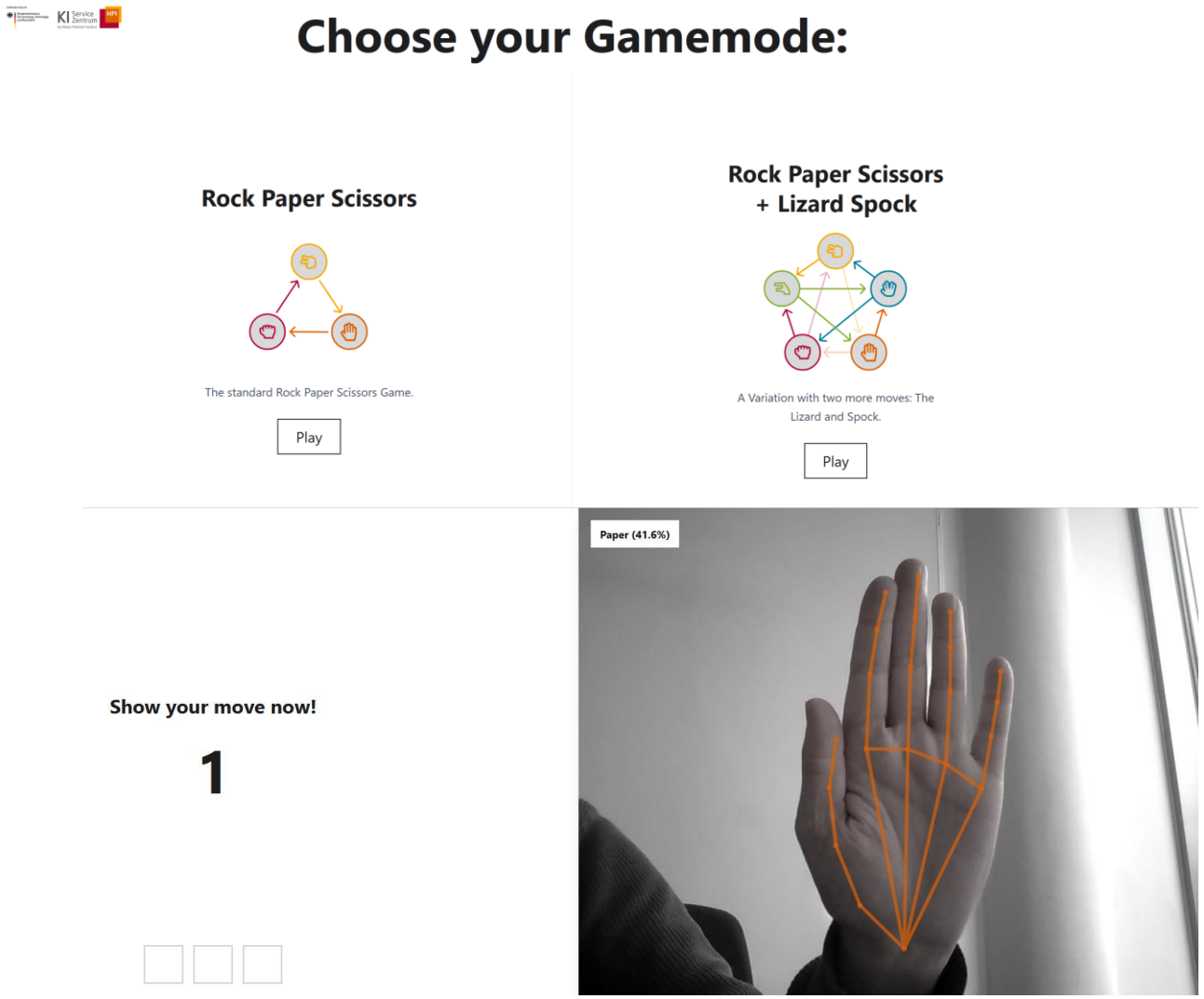



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# DEMONSTRATOR: VISIONARY GESTURES

https://vadimitri.github.io/visionarygestures/

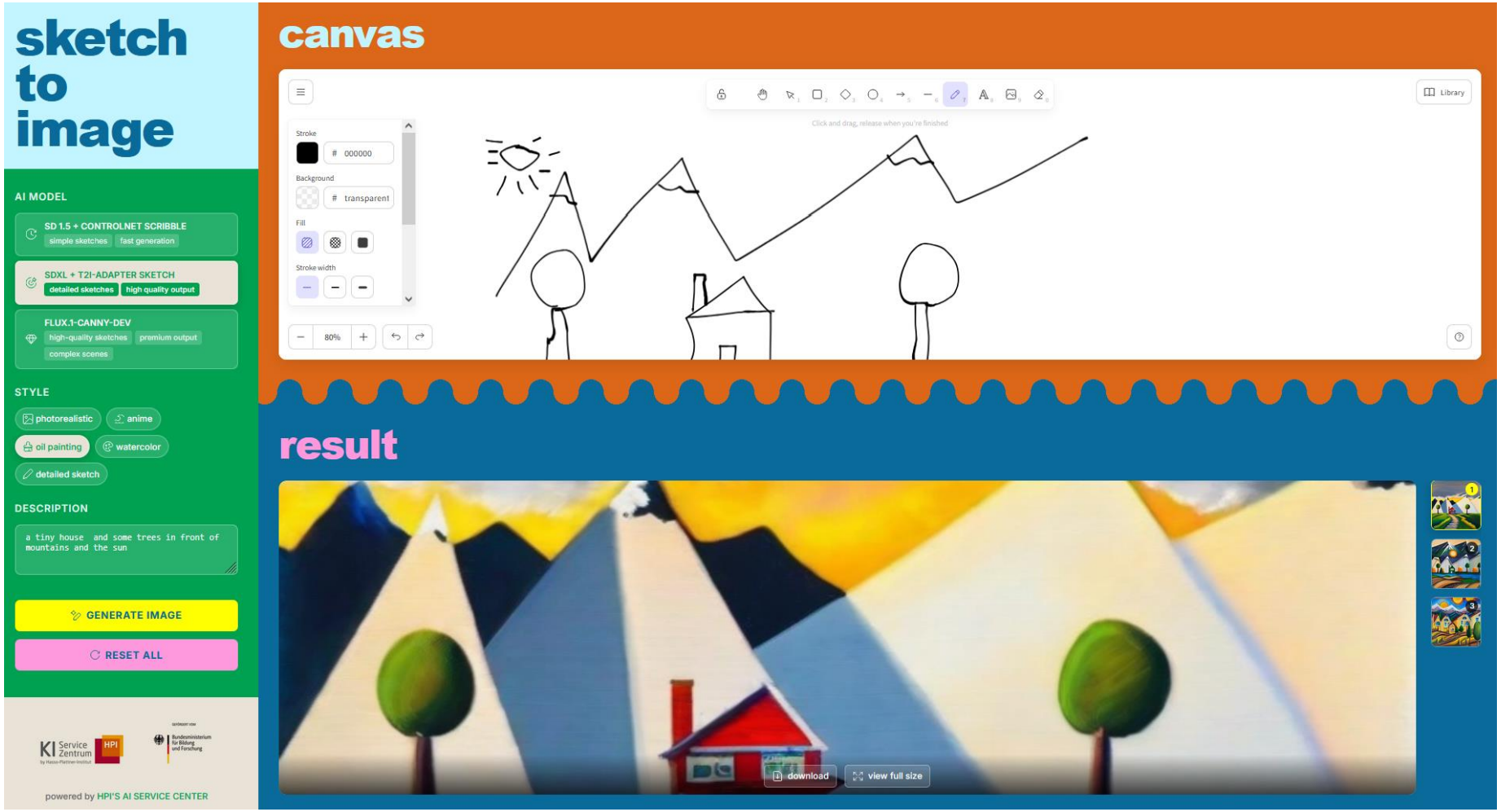



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# DEMONSTRATOR: SKETCH2IMAGE

https://s2i.aisc.hpi.de/
user: aisctester
pw: 9OzJ5TzJZybVjm




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KI Service Zentrum by Hasso-Plattner-Institut HPI

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Email:
kisz@hpi.de

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ZUM SHOWROOM

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# Your feedback is important!

QR-Code for feedback questionaire

